

NANYANG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE
NAME

Mark Scheme

CLASS

BIOLOGY

Paper 2 Core Paper

9648/02

September 2013

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your name and CT on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer any **one** question.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use

Section A	
1	
2	
3	
4	
5	
6	
7	
Section B	
Total	

This document consists of **18** printed pages and **no** blank page.

[Turn over

Section A

Answer **all** the questions in this section.

- 1 The electron micrograph in **Fig. 1.1** shows an acinar cell found in the pancreas.

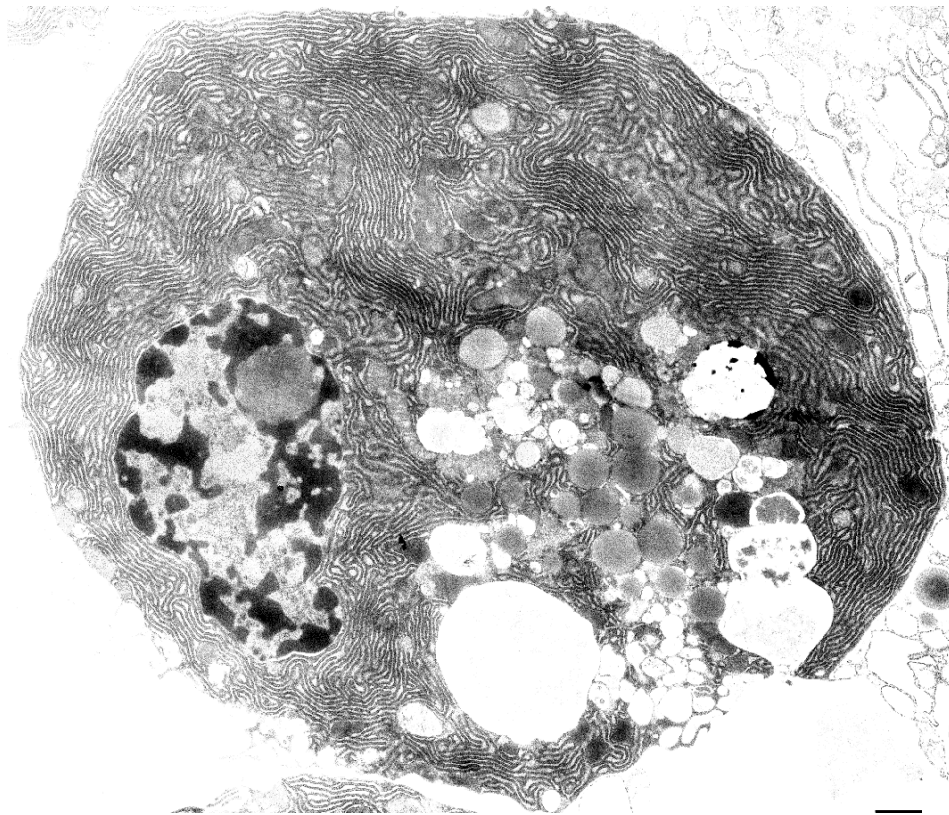


Fig. 1.1

- (a) With reference to the **Fig. 1.1**, suggest the principal function of the pancreatic acinar cell. Give **two** reasons to explain your answer.

secretion of pancreatic / digestive enzymes;

.....
 ANY TWO

presence of large amount of rER, which is the site where pancreatic enzymes are synthesized;

presence of large number of secretory vesicles for exocytosis to release of pancreatic enzymes outside the cell;

presence of mitochondria which synthesizes ATP for activation of amino acids / Formation of peptide bonds (at ribosomes) / movement of vesicles towards cell surface membrane for exocytosis of pancreatic enzymes ;

[3]

Fig. 1.2 shows two different structures, **A** and **B**, found in an animal cell.

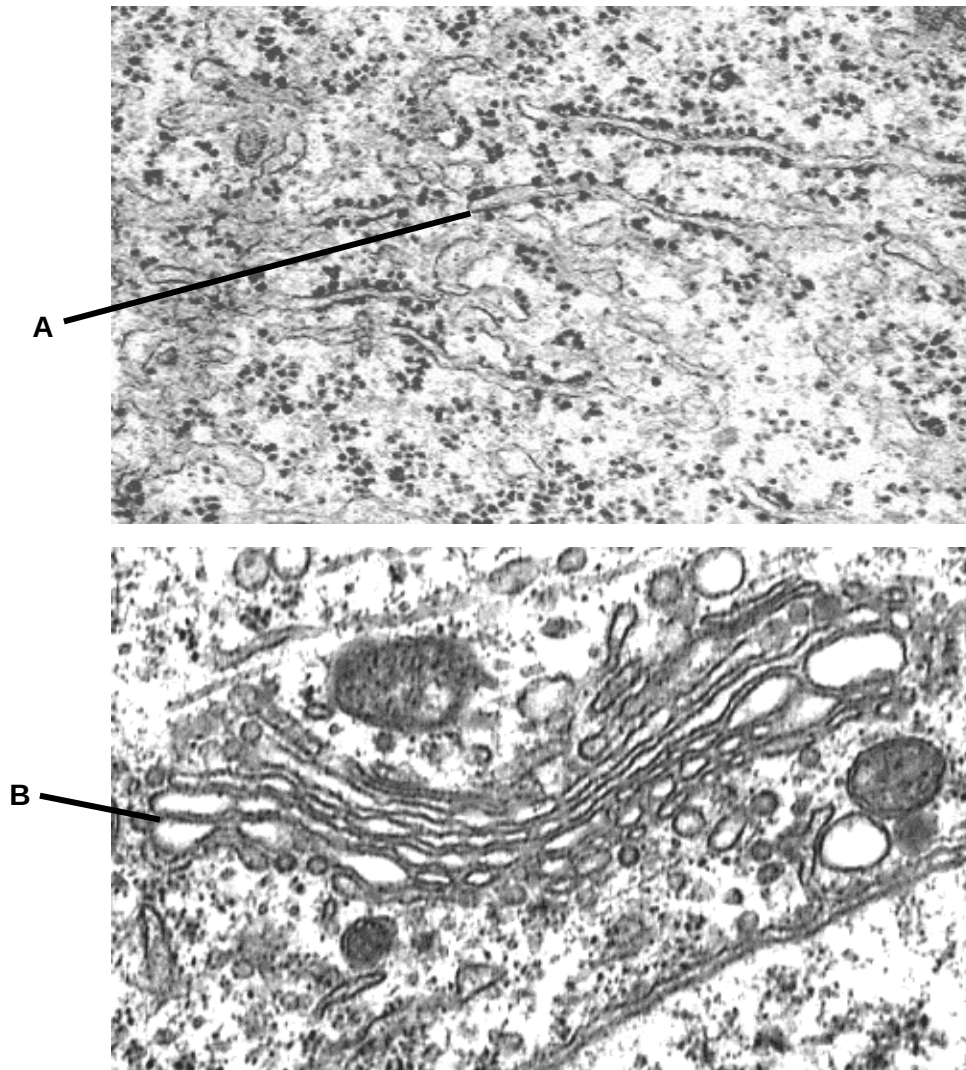


Fig. 1.2

(b) With reference to the Fig. 1.2, describe **three** ways in which **A** is similar to **B**.

Both are bound by a single membrane / single membrane-bound structures.

Both consist of membranous sacs called cisternae, in which newly synthesized proteins undergo processing.

Both give rise to membrane-bound vesicles which are involved in transport of newly synthesized proteins within the cell.

[3]

Scientists isolated mitochondria from the liver cells. They broke the cells open in an ice-cold, isotonic solution. They then used a centrifuge to separate the cell organelles. **Fig. 1.3** illustrates the steps that take place in the process of centrifugation.

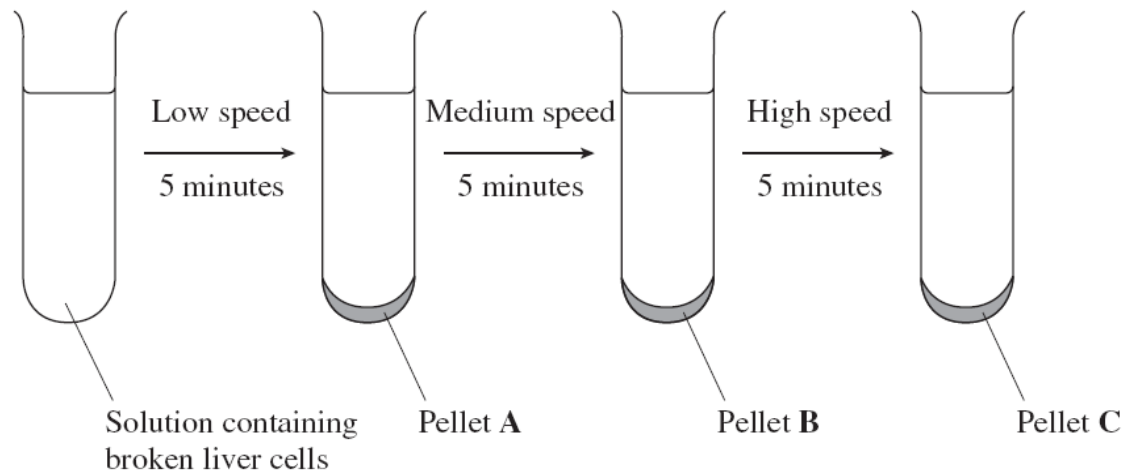


Fig. 1.3

(c) State and explain which pellet contains mitochondria.

Pellet B ;

Mitochondria is the next biggest organelle after the nucleus ;

[2]

(d) Suggest why the solution used was ice-cold and isotonic.

Ice cold is to reduce enzyme activity / cellular preservation;

Isotonic to prevent net movement of water so organelle will not burst or shrivel / maintain organelle's integrity;

[2]

People with mitochondrial disease have mitochondria that do not function properly. These disorders may be caused by mutations in the mitochondrial DNA or in the nuclear genes that code for mitochondrial components.

(e) Explain why a person with mitochondrial disease can only exercise for a short time.

Mitochondria use aerobic respiration to synthesize ATP ;

ATP is needed for muscular contractions ;

[2]

[Total: 12]

- 2 Prokaryotes were first discovered in the late 1600s by Antony van Leeuwenhoek, using the microscope he invented. With the advancements in science, scientists have been able to unravel the mysteries between prokaryotes and eukaryotes.

Fig. 2.1 (a) and (b) shows protein synthesis carried out by the two types of cells.

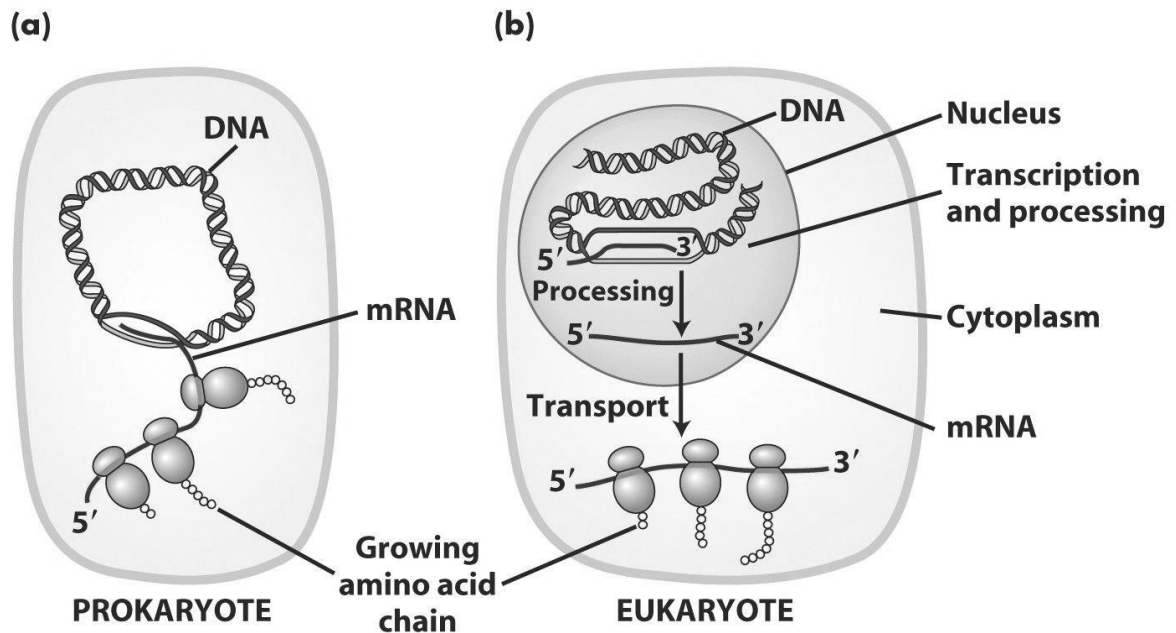


Fig. 2.1

- (a)** Describe the structure of a RNA molecule that is not shown in **Fig. 2.1** in relation to its function.

Transfer RNA is a single-stranded RNA /, which is folded into a clover leaf structure then into its three-dimensional structure + to bind to ribosome/ to fit into EPA site of ribosome;

tRNA possess an anticodon which base pairs complementarily with codon on mRNA;

tRNA has an acceptor stem / 3' end for amino acid attachment, thus allowing tRNA to transfer the amino acid to the ribosome during translation;

@ rRNA is single stranded and a component of ribosomes;

[3]

- (b)** Describe the differences between prokaryotes and eukaryotes with respect to gene expression at the following levels:

- (i)** Chromatin level

In eukaryotes, histone acetylation neutralizes positive charge and DNA more accessible to RNA polymerase for transcription

whereas in prokaryotes, packing of prokaryotic DNA do not involve histone proteins / histone modification does not occur;

In eukaryotes, DNA methylation attracts enzymes which increase condensation and decreased rate of transcription resulting in gene silencing

whereas in prokaryotes, DNA methylation discriminates between foreign and endogenous DNA for restriction endonucleases;

[2]

(ii) Transcriptional level

In eukaryotic, TATA binding proteins/Transcription factors binds to TATA box/promoter recruiting RNA polymerase and basal TF for the assembly of transcription initiation complex

In prokaryotes: Prokaryotic RNA polymerase can bind directly to core promoter without aid of TF, resulting in basal rate of transcription of operon

In eukaryotes, binding of activators to enhancers loop the DNA and increases rate of transcription

In prokaryotes, usually lack distal control elements but contains CAP site for binding of CAP and increases rate of transcription

In eukaryotes, binding of repressors to silencer decreases rate of transcription

In prokaryotes, binding of repressor to operator site within operon prevents transcription allowing inducibility of operon

[3]

During research into the mechanism of DNA replication, bacteria were grown for many generations in a medium containing only the 'heavy' isotope of nitrogen, ^{15}N . This resulted in all the DNA molecules containing only ^{15}N . This is illustrated in **Fig. 2.2** below.

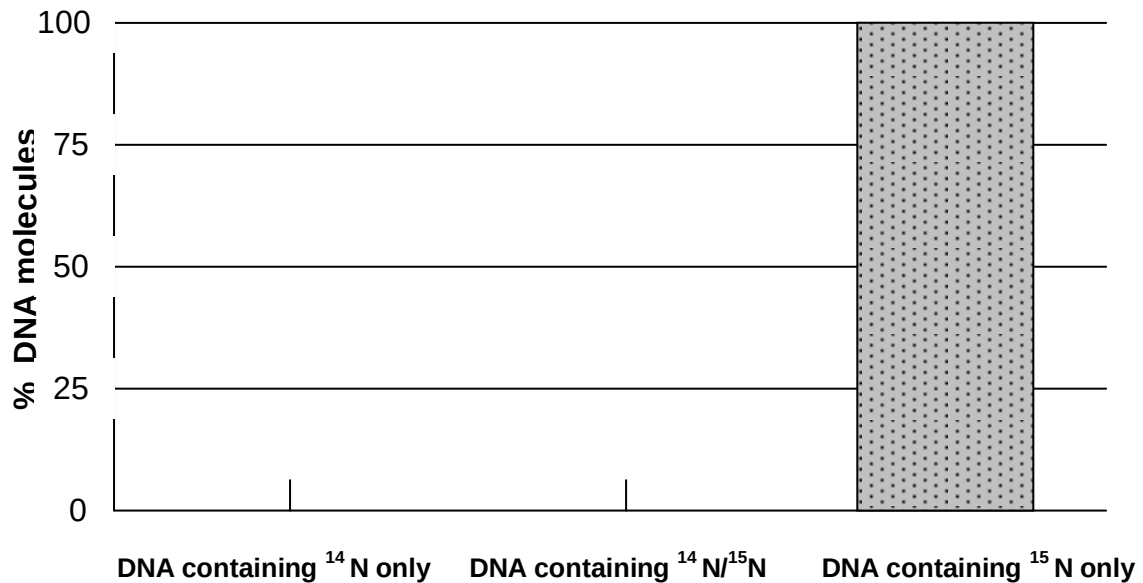


Fig. 2.2

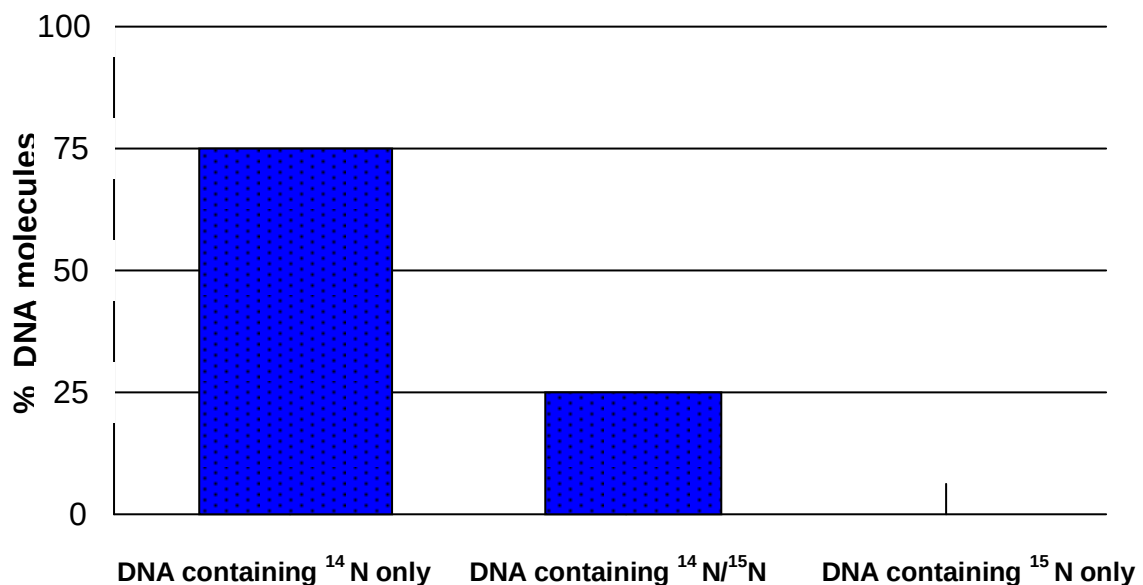
Step 1: These bacteria were then grown in a medium containing only 'light' nitrogen, ^{14}N . After the time taken for the DNA to replicate once, the DNA was analysed.

Step 2: The bacteria continued to grow in the 'light' nitrogen, ^{14}N , medium until the DNA had replicated once more. The DNA molecules were then analysed.

Step 3: The bacteria was allowed to grow in the 'light' nitrogen, ^{14}N , medium for another round of DNA replication.

(c) Complete the bar chart below to indicate the expected results of the composition of these DNA molecules after **step 3**.

[1]



DNA replication is a process which involves DNA polymerase and the deoxyribonucleotides A, T, C and G, while transcription involves RNA polymerase and the ribonucleotides A, U, C and G.

(d) Describe two **other** differences and two similarities between DNA replication and transcription.

.....

.....

.....

.....

.....

.....

[4]

[Total: 13]

Differences

Features	DNA replication	Transcription
Template	<u>Both strands</u> of the double helix are used as templates.	Only <u>one strand</u> of the double helix is used as template.
Parts of template copied	<u>Entire DNA molecule / Entire chromosome / Entire or All of 2 strands</u> is/are replicated	Only <u>short segments</u> of DNA / <u>gene</u> is transcribed to form RNA
Product	<u>Double stranded</u> DNA	<u>Single stranded</u> RNA which can be mRNA, tRNA or rRNA

Similarities

- Both involve unwinding and unzipping double helix DNA/ separation of 2 DNA strands by breaking of the weak hydrogen bonds between complementary bases
- Both involve formation of phosphodiester bonds between neighbouring nucleotides
- Both involve aligning free nucleoside triphosphates / nucleotides through complementary base pairing
- Both products elongate in 5' to 3' direction
- Both require energy from ATP hydrolysis

- 3 **Fig. 3.1** is an electron micrograph showing T4 bacteriophages invading a bacterium.

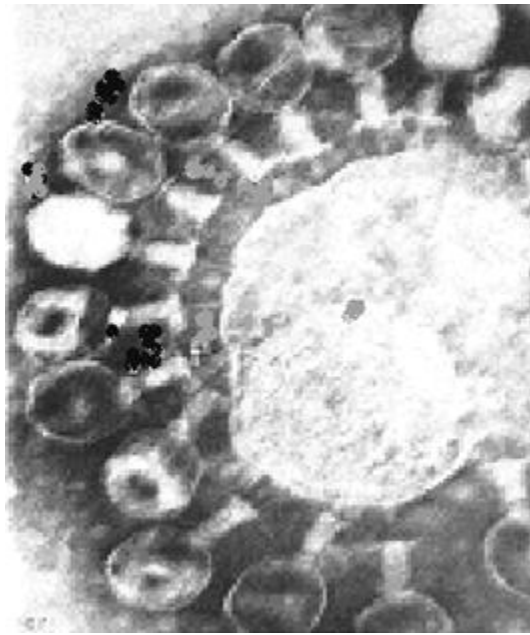


Fig. 3.1

- (a) Clearly label the base plate of one T4 phage in **Fig. 3.1**. [1]

Answer must show clear line pointing to base plate with appropriate label ("base plate" or "(a)").

- (b) Explain how T4 may mediate the transfer of a random piece of this bacteria's genetic material to another bacteria.

In generalized transduction;

After T4 DNA entered the donor bacterium, phage enzymes hydrolyze bacteria DNA;

During assembly of new phage virus, random piece of hydrolysed bacterial DNA packaged within phage capsid;

After lysis of bacterial cell, phage infects recipient bacterium, injecting donor DNA along with its own;

Homologous recombination may result in recipient bacteria acquiring new characteristics;

® Specialised transduction because phage is T4 and genetic material is "random piece of bacterial genetic material".

[4]

- (c) Briefly describe the structure of bacterial chromosomes.

1 double-stranded circular DNA;

Form looped domains with small amount of DNA-binding proteins; OR

Further supercoiled to form chromosome;

[2]

Some genes in bacteria are organized into operons, such as the *lac* operon. Transcription of the *lac* operon in *E. coli* is dependent on the presence or absence of food molecules such as glucose and lactose.

- (d) **Under normal conditions**, predict the transcription of the operon in the presence or absence of these two substrate molecules by writing “**High**” or “**Low**” in the table below.

Lactose	Glucose	Transcription
+	+	Low
+	-	High
-	+	Low
-	-	Low

[1]

- (e) **Mutation XX-6** deletes the DNA binding site from the CAP protein, but leaves the rest of the operon unaffected. Predict the effect on transcription under these four conditions by writing “**High**” or “**Low**” in the table below.

Lactose	Glucose	Transcription
+	+	Low
+	-	Low
-	+	Low
-	-	Low

[1]

- (f) Another mutation on a separate site gives the following results.

Lactose	Glucose	Transcription
+	+	Low
+	-	High
-	+	Low
-	-	High

Suggest the mutation that could have occurred and explain how it would have given rise to the above results.

Mutation of *lacI* gene, resulting in non-functional repressor / repressor that no longer binds to operator;

OR

Mutation of operator such that repressor can no longer bind (without affecting binding of RNA polymerase to promoter);

RNA polymerase free to bind to promoter to initiate transcription regardless of levels of glucose present.

[2]

[Total: 11]

- 4 A woman's risk of developing breast cancer is greatly increased if she inherits *BRCA1* gene mutation. The role of BRCA1 protein is shown in **Fig. 4.1**.

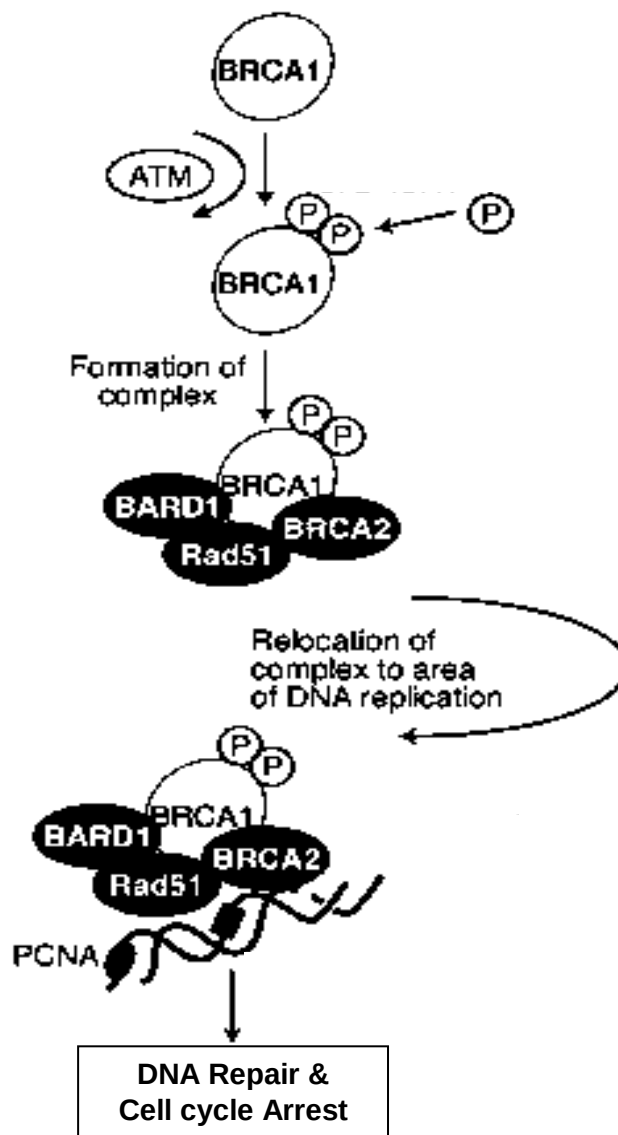


Fig. 4.1

- (a) With reference to **Fig. 4.1**,
 (i) State the class of genes that *BRCA1* belongs to.

[Tumour suppressor genes;](#)

[1]

(ii) Explain how mutations in *BRCA1* gene can contribute to cancer.

Loss of function mutation in *BRCA1* gene which codes for proteins involved in DNA repair / repair damaged DNA;

DNA repair do not occur resulting in accumulations of mutations in proto-oncogene / tumour suppressor genes / telomerase gene; OR

Resulting in uncontrolled cell division forming a tumour; OR

Resulting in tumour angiogenesis, invasion of cancer cells into surrounding tissues and metastasis to distant sites;

[3]

(iii) The expression of *BRCA1* gene can be furthered controlled at post-translational level. Explain how this can occur.

Biochemical modification by phosphorylation / addition of phosphate groups to *BRCA1* protein by ATM kinase;

Phosphorylated *BRCA1* protein is functional and able to form protein complex / OWTTE (effect of phosphorylation);

[2]

(b) The *HER2* gene codes for a growth factor receptor. When the growth factor binds to *HER2* protein (the growth factor receptor), the signal is given for the cell to start dividing. Gene amplification of *HER2* gene could also result in breast cancer.

(i) Describe a process which results in gene amplification.

Any 1 pair

During DNA replication, DNA polymerase may slip from the DNA template strand;

then reattach and replicate a region that was already replicated;

OR

A copy of the repeat sequence in one chromosome can misalign with a different copy of the repeat in the homologous chromosome;

Unequal crossing over occurs, resulting in one chromosome being longer than the other;

[2]

(ii) Suggest how amplification of *HER2* could result in breast cancer.

Increases the number of copies of *HER2* gene / causes over-expression of *HER2* growth receptors;

Therefore more sensitive to growth factors and thus stimulate uncontrolled cell division;

[2]

[Total: 10]

- 5 Rickets is a childhood disorder involving the softening and weakening of bones in humans. It is usually caused by a lack of vitamin D, calcium ions or phosphate ions. A rare form of rickets that cannot be successfully treated with vitamin D therapy is caused by an allele on the X chromosome that arose as a result of a gene mutation.

(a) Explain what is meant by the term gene mutation.

Refers to a change in DNA / base / nucleotide sequence;

e.g. addition / substitution / deletion of a nucleotide / a base pair;

no / different protein / polypeptide is produced;

produces a different allele;

[2]

(b) Fig. 5.1 below illustrates how this rare form of rickets is inherited in an afflicted family.

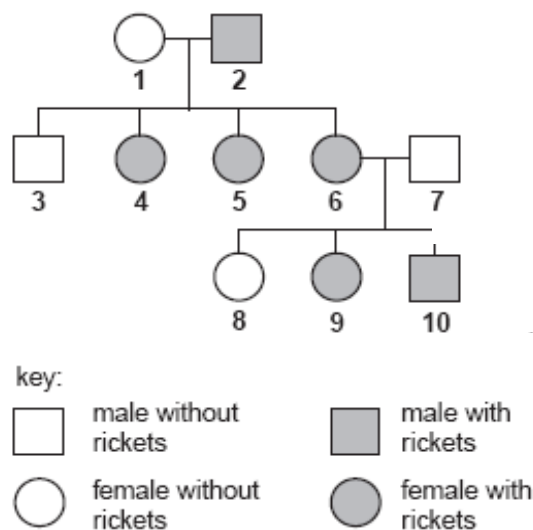


Fig. 5.1

- (i) With reference to Fig. 5.1, state and explain the pattern of inheritance of this form of rickets.

(Sex-linked) dominant condition;

Individual 2 has one X chromosome with dominant mutant allele and will pass to all the daughters 4, 5 and 6;

OR Individual 6 can pass the dominant mutant allele to their daughter 9;

The dominant mutant allele will mask the effect ® expression of the recessive normal allele / only one copy of the mutant allele required to manifest rickets;

[3]

- (ii) Using appropriate symbols, state the genotypes of the following individuals.

1 $X^r X^r$

3 $X^r Y$

9 $X^R X^r$

10 $X^R Y$

[2]

(iii) Couple 6 and 7 decided to have another son. Draw a genetic diagram to determine the probability that their second son will suffer from rickets.

Parental phenotypes:

Normal male

Female with rickets

Parental genotypes:

 $X^r Y$ $X^R X^r$

;

After meiosis,
gametes produced X^r Y X^R X^r

By random fertilisation,

	X^r	Y
X^R	$X^R X^r$	$X^R Y$
X^r	$X^r X^r$	$X^r Y$

;

offspring genotypes:

 $X^R X^r$ $X^r X^r$ $X^R Y$ $X^r Y$

offspring phenotypes:

1 Female
with rickets: 1 Female
without rickets: 1 Male
with rickets: 1 Male
without rickets

1/2 or 50% of the sons will suffer from rickets;

- (c) Colour blindness and haemophilia are inherited sex-linked recessive conditions in humans. **Fig. 5.2** shows two generations of a family in which these conditions are inherited.

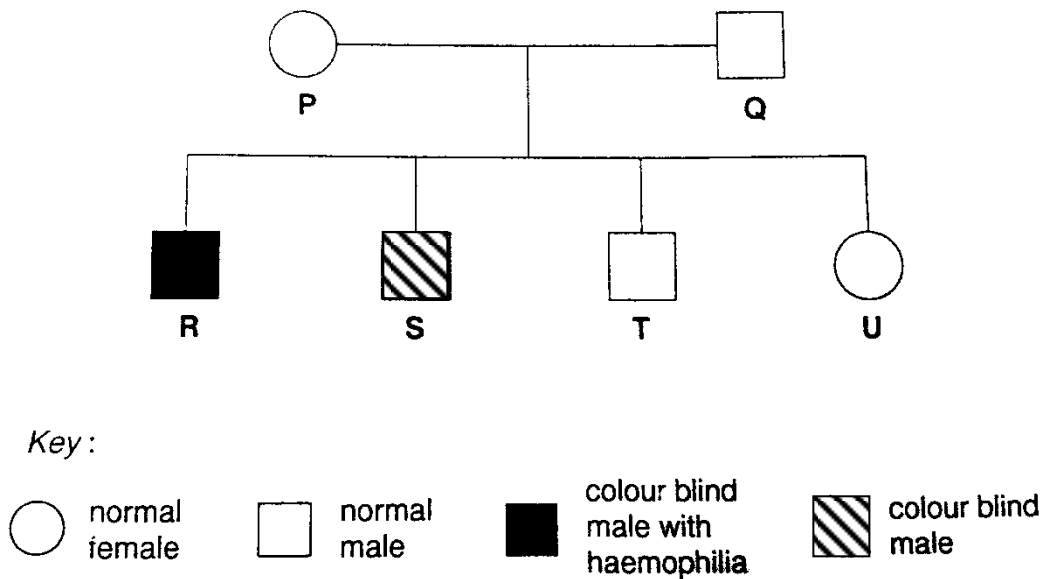


Fig. 5.2

Suggest how the genotype of **S** has arisen.

Crossing over of non-sister chromatids of homologous chromosomes during Prophase I of meiosis in P;

Exchange of alleles between X chromosomes and S inherit a X chromosome carrying the (dominant) normal allele for haemophilia and (recessive) mutant allele for colour blind;

Both genes are linked / found on the X chromosome;

[2]

[Total: 12]

- 6 **Fig. 6.1** below shows the cell signalling pathway in smooth muscle cells in response to the hormone Epinephrine, resulting in smooth muscle relaxation of blood vessels (vasodilation).

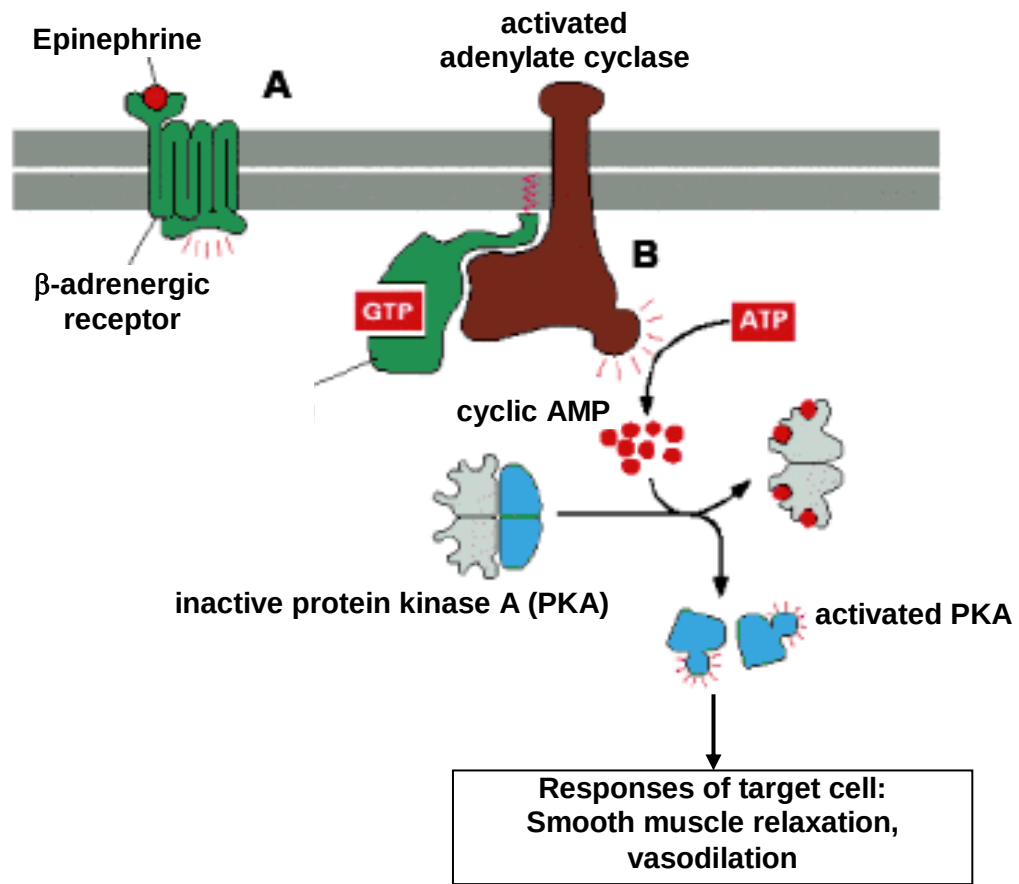


Fig. 6.1

- (a) State the class of receptors that the β -adrenergic receptor belongs to.

G protein coupled receptors / G protein linked receptors;

[1]

- (b) With reference to **Fig. 6.1**,

- (i) describe how adenylate cyclase is activated.

When the epinephrine binds to the β -adrenergic receptor, a ligand-receptor complex is formed causing the receptor undergoes a conformation change;

Bound G-protein (at the cytoplasmic) end of the receptor to exchange its GDP with a GTP / substitute GTP for GDP;

the G-protein is now activated and translocates / moves along the membrane to bind to and activate adenylate cyclase

[3]

- (ii) identify the stage where signal amplification is observed and explain how the signal is amplified in that stage.

Stage B / Conversion of ATP to cAMP by adenylate cyclase;

Each enzyme produces many molecules of cAMP;

[2]

- (c) Explain how the structure of the β -adrenergic receptor allows it to carry out its function as a receptor.

Transmembrane protein embedded on cell surface membrane;

Extracellular binding site is complementary/specific to ligand / epinephrine;

Intracellular domain able to bind to inactive G-protein / able to undergo conformational change to cause activation of G-protein;

[3]

- (d) A drug was designed to ensure that permanent vasodilation occurs. Suggest how this drug might work within the cell signalling pathway in **Fig. 6.1**.

Similar structure to epinephrine and binds permanently to β -adrenergic receptor;

G-protein permanently activated, adenylate cyclase permanently activated, continuously synthesizes cAMP;

[2]

[Total: 11]

- 7 Scientists studying the evolutionary history of the Human Immunodeficiency Virus (HIV) found that it is closely related to the Simian Immunodeficiency Virus (SIV) found in certain species of monkeys. The phylogenetic tree in **Fig. 7.1** shows part of the evolutionary relationships between these viruses. A common ancestral virus is believed to have evolved into strains that infected chimpanzees (SIV). Over time, new strains emerged which infect humans.

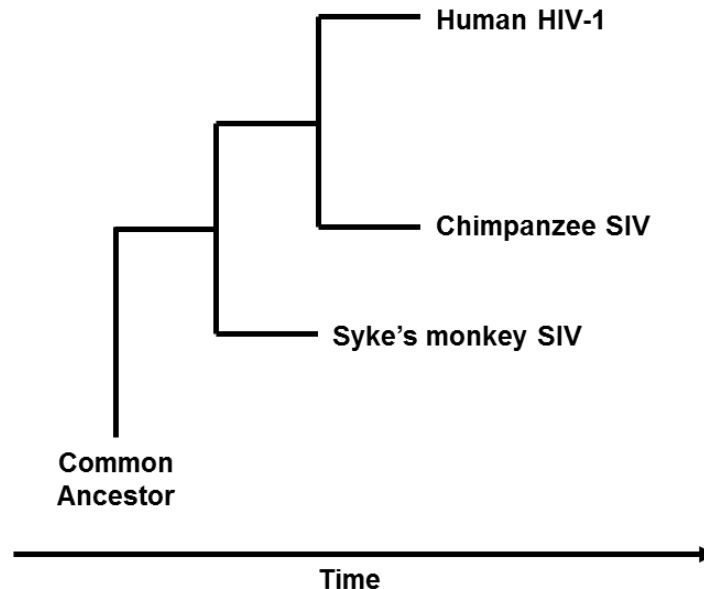


Fig. 7.1

- (a) Explain the significance of the branch points in the phylogenetic tree shown in **Fig. 7.1**.

Each branch point represents the common ancestor shared by descendants / divergence of two species from common ancestor;

Position of branch point corresponds to the relative time when two species had diverged;

[2]

- (b) Explain how molecular homology could be used to support the claim that HIV and SIV share an evolutionary relationship.

Direct comparison of DNA / RNA of a homologous sequence / amino acid sequence of a common gene;

Two viral species that show high similarity in their sequences probably share a common ancestor;

The more similar the sequences of two species are, the more closely related they are in their evolutionary relationship.

[2]

(c) State **two** advantages of using molecular techniques in classification.

Degree of divergence between different species can be quantitatively measured by comparison of nucleotide / amino acid sequences;

This method is objective compared to comparison of morphology which may be subjective;

Evolutionary changes between species can be compared even though they may differ vastly in terms of morphology / Molecular comparisons transcend barriers among organisms whose relationships cannot be evaluated by traditional experimental techniques;

[2]

SIV infections in their natural hosts appear in many cases to be non-pathogenic (non-disease causing). Certain individuals within the human population have also shown similar characteristics when exposed to HIV. These individuals continue to test negative for HIV despite being repeatedly exposed to HIV.

Investigations have found a mutation present in the CCR5 gene in these individuals. CCR5 is a co-receptor that facilitates the entry of HIV into the T-cell after gp120 (present on the viral envelope) binds with the CD4 receptor.

(d) Suggest and explain how these individuals continue to resist HIV infection.

A CCR5 gene mutation may result in lowered expression of CCR5 protein / change in structure of CCR5 / non-functional CCR5 receptor on cell surface membrane of helper T-cells;

Virus envelope unable to fuse with cell surface membrane of helper T-cells and thus unable to replicate in T-cells;

[2]

- (e) Assuming that humans do not take active steps to prevent HIV infection, explain how the human population could possibly evolve to not be affected by HIV.

Variation in genotype of humans some with CCR5 mutation, some without;

@ HIV resistant individuals and non HIV resistant individuals

® individuals with HIV and individual without HIV

.....
When exposed to HIV, humans with CCR5 mutation that prevent entry of HIV are able to survive and thus possess a selective advantage over other humans without CCR5 mutation;

.....
Thus this favourable allele ® trait is inherited by the offspring;

.....
Allele frequency for CCR5 mutation increases in human population, over long period of time, most humans will be unaffected by HIV;

.....
[3]

[Total: 11]

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 8 (a) Outline how the endomembrane system is involved in synaptic transmission. [8]
 (b) Describe the main stages of the Calvin cycle [8]
 (c) Contrast the structure of cellulose with that of collagen and suggest why plant cells have cellulose but not collagen. [4]
- 9 (a) Compare and contrast the structures of influenza and HIV. [8]
 (b) Describe how HIV infects host cells and reproduces. [8]
 (c) Briefly explain how a person infected only **once** by one strain of HIV may eventually have many strains of HIV. [4]

(a) Outline how the endomembrane system is involved in synaptic transmission. [8]

- 1 Synaptic transmission involves membrane proteins such as voltage-gated Ca^{2+} channels in the pre-synaptic membrane / ligand-gated ion channels in the post-synaptic membrane.
- 2 Both type of channels are integral membrane protein which is first translated by ribosomes attached to rER;
- 3 and modified / folded into right conformation in the rER lumen;
- 4 The folded proteins are then transported to the cis face of Golgi apparatus via transport vesicles which bud off the rER;
- 5 When moving from cis to trans face of the Golgi apparatus, the proteins are further chemically modified such as glycosylation, phosphorylation, dehydration, and concentration.
- 6 The protein is then inserted to the membrane of secretory vesicles which bud off the trans face of Golgi apparatus;
- 7 The membrane of secretory vesicles move towards and fuse with the cell surface membrane, incorporating the channel proteins in the membrane;
- 8 Some neurotransmitters of peptide nature are also synthesized and packaged in the same way for exocytosis. (Note: Acetylcholine is NOT a peptide hormone)
- 9 However, they are enclosed in vesicles instead of inserted into the vesicle membrane.
- 10 Acetylcholinesterase embedded on post-synaptic membrane also synthesized in similar manner to channel proteins.

(b) Describe the main stages of the Calvin cycle.

[8]

- 1 Calvin cycle occurs in the stroma of chloroplasts;
- 2 During CO₂ fixation, ribulose biphosphate (RuBP) accepts CO₂ to form an unstable 6C intermediate;
- 3 catalyzed by ribulose biphosphate carboxylase-oxygenase (Rubisco);
(Note that rubisco is NOT an acceptable abbreviation)
- 4 Unstable 6C intermediate immediately breaks down to 2 molecules of 3C compound known as glycerate phosphate / phosphoglycerate (GP / PGA);
- 5 GP can be converted to pyruvate which is used to synthesize fatty acids/amino acids;
- 6 During reduction, GP is then converted to triose phosphate / glyceraldehyde phosphate (TP / GALP)
- 7 using energy from hydrolysis of ATP and reducing power from reduced NADP, produced by light dependent reactions;
- 8 TP contains more chemical energy than PGA / GP, and is the first carbohydrate made in photosynthesis to leave the Calvin cycle;
- 9 5/6 of the total amount of TP is used to regenerate the RuBP consumed in the first reaction which requires energy from the hydrolysis of ATP;
- 10 1/6 of the total amount of TP is used to make other carbohydrates and glycerol;
- 11 6 rounds of the cycle is required to produce 1 molecule of glucose;

(c) Contrast the structure of cellulose with that of collagen and suggest why plant cells have cellulose but not collagen. [4]

- 1 Cellulose is a polysaccharide whereas collagen is a fibrous protein;
- 2 Monomers which make up cellulose are β -glucose residues whereas monomers that make up collagen are amino acids (® if only mention glycine/proline/hydroxyproline. Other amino acids are also present albeit in much lower numbers!);
- 3 β -glucose in cellulose are linked by β -1,4-glycosidic bonds whereas amino acids in collagen are linked by peptide bonds;
- 4 β -glucose in cellulose are rotated 180° relative to each other such that they form straight chains whereas collagen is formed by coiling 3 helical polypeptides around each other to form a tropocollagen;
- 5 Genes encoding for polypeptides of collagen is absent / not expressed in plant genome;
- 6 Lack enzymes necessary for the post-translational processing / hydroxylation of proline / glycosylation of amino acids in collagen;
- 7 Plant cells can undergo photosynthesis, so glucose is readily available for the synthesis of cellulose whereas amino acids are not as readily abundant for formation of collagen;

(a) Compare and contrast the structures of influenza and HIV. [8]

- 1 Both have envelopes with embedded glycoproteins;
- 2 Envelopes of both are derived from host cell surface membrane;
- 3 Both genomes are single-stranded RNA;
- 4 Both are spherical in shapes;

Features	Influenza	HIV
5. Genome;	Single-stranded, segmented <u>negative sense</u> RNA	Single-stranded <u>positive sense</u> RNA
6.;	which is <u>complementary</u> in sequence to <u>mRNA</u>	which is <u>identical</u> in sequence to <u>mRNA</u>
7. Number of copies of genome;	<u>1 copy segmented into 8 pieces</u>	<u>2 identical</u> copies
8. Types of glycoprotein (receptor) ;	Haemagglutinin and neuraminidase	<u>gp120</u> and gp 41
9. Shape of capsid;	<u>Helical</u> nucleoprotein	<u>Conical</u>
10. Enzymes present;	Viral RNA-dependent RNA polymerase	Integrase, protease and reverse transcriptase

(b) Describe how HIV infects host cells and reproduces. [8]

Adsorption / Attachment

- 1 gp120 on viral envelope recognize and bind to specific receptor molecules (CD4⁺) on cell surface membrane of T helper cells of immune system;

Entry / Penetration

- 2 Viral envelope fuses with cell surface membrane;
- 3 Capsid is digested by cellular enzymes (® exonucleases), releasing the viral RNA molecules, viral proteins and enzymes into the cytoplasm.

Integration

- 4 Reverse transcriptase catalyses the synthesis of a single DNA strand complementary to the viral RNA;
 - 5 The viral RNA is degraded and reverse transcriptase catalyses the synthesis of a second DNA strand complementary to the first;
- (OR
4 & 5 may be combined for a single mark as:
Reverse transcriptase catalyses the synthesis of double stranded DNA from RNA template)
- 6 The double-stranded viral DNA then enters the cell's nucleus and integrates, as a provirus, into the host cell DNA via the action of integrase;
(The provirus never leaves the host's genome, remaining permanently in the host cell.)

Synthesis of viral components

- 7 When host cell is activated, proviral genes are transcribed into RNA;
- 8 RNA is used as the HIV genome;
- 9 Or serves as mRNA which is translated into viral proteins (capsid protein and viral glycoproteins at ribosomes in cytoplasm and surface of endoplasmic reticulum respectively);
- 10 Golgi vesicles embedded with viral glycoproteins migrate towards and fuse with the cell surface membrane such that viral glycoproteins become embedded on the cell surface membrane;
- 11 Polyproteins are similarly cleaved by proteases to produce structural proteins.

Viral assembly / maturation

- 12 Capsid proteins enclose viral genome and viral proteins;

Release

- 13 Virus buds off from the cell (®exocytosis), surrounded by host cell surface membrane embedded with viral glycoproteins;

(c) Although a person may be infected only once by one strain of HIV he may eventually have many strains of HIV. Briefly explain why. [4]

- 1 Different strains of HIV are genetic variation that arise via mutations;
- 2 HIV mutates very rapidly;
- 3 Reverse transcriptase is inaccurate in its replication / Unlike DNA polymerase, reverse transcriptase does not proof read the nucleotides added;
- 4 Reproductive rate of HIV is high → increases chances of mutant strains arising;
- 5 Mutation rate is greater because the genetic material is single stranded (RNA);
- 6 RNA single stranded → less stable because no complementary strand → no template for DNA repair;